

30.103 Fluid Mechanics Design Project (1D Project)

2026 EPD Term 5

Instructor: Prof. Michinao Hashimoto

Title: Vibe Coding to Develop Interactive Engineering Simulators

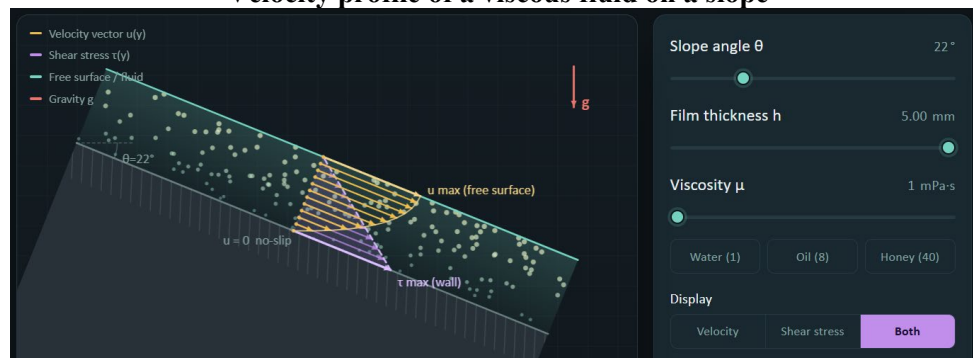
Background

In traditional engineering workflows, translating complex physical theorems into working numerical simulations requires extensive programming expertise. **Vibe Coding** shifts this paradigm by using natural language and generative artificial intelligence (such as Claude, Gemini, or ChatGPT) to write functional code efficiently. This approach enables engineers and students to rapidly prototype interactive tools without being held back by a lack of coding skills.

Nevertheless, the foundation of any engineering subject, including fluid mechanics, relies on fundamental governing equations. Governing equations are not generated by AI; rather, they are rooted in established first principles of physics. As we learn and derive these various governing equations of fluid mechanics in class, the goal of this project is to leverage Vibe Coding to build simulators that directly reflect these first principles. By prompting an AI with the underlying physics, boundary conditions, and fluid constraints, you can generate visual tools to explore complex flow behaviours dynamically.

Examples here: <https://michinaoh.github.io/fluid-simulation/>

Velocity profile of a viscous fluid on a slope



Bernoulli's Principle



Learning Objectives

- **Understand** the concept of Vibe Coding and its application in rapid engineering prototyping.
- **Translate** fundamental governing equations of fluid mechanics (e.g., Navier-Stokes, Bernoulli, and hydrostatic equations) into structured natural language prompts for code generation.
- **Implement** an iterative prompt-refinement workflow to add variables, boundary constraints, and physical phenomena to a simulation step by step.
- **Design** an intuitive Graphical User Interface (GUI) that enables interactive parameter manipulation and real-time flow visualization (e.g., streamlines, particle tracking).
- **Apply** AI-generated simulators to deepen the understanding of classroom concepts, solve complex engineering problems, or inform physical device designs.

Project Aims

Through this 1D design project, students will:

- **Formulate and code** an interactive, web-based or GUI-driven simulator designed to dynamically visualize core fluid mechanics phenomena.
- **Iterate and expand** the mathematical model from a simplified baseline into a complex system by systematically injecting new physical variables, geometric variations, and boundary constraints.
- **Develop** a responsive GUI that enables users to execute real-time parameter sweeps (e.g., varying velocity, dimensions, elevation) and immediately observe the fluid system's response.
- **Validate** the simulator's quantitative outputs by comparing them directly against classical analytical solutions and classroom first principles.

Examples of Simulator Topics

Students may select any core topic covered in the 30.103 Fluid Mechanics syllabus, including but not limited to:

- **Venturi Effect and Bernoulli Principle:** Simulators incorporating throat constrictions, elevation variations, and cavitation thresholds.
- **Pipe Flow Mechanics:** Interactive simulator modelling friction factors, major/minor head losses, and pressure drops.
- **Laminar/Turbulent Flows:** Visualizations of laminar versus turbulent flows and Reynolds number transitions across varying channel geometries.
- **Control Volume Analysis:** Simulators mapping fluid momentum, forces on pipe bends, or conservation of mass.

Alternative Approach can be to generalize homework and case problems. You are highly encouraged to build an interactive solver based on an existing homework or case problem from class. Rather than simply hardcoding a single solution (with the given numerical values), your simulator can **generalize the problem and make them more complex**. It should feature full parameter variation (allowing the user to dynamically adjust geometry and fluid properties) to help others gain a deeper, intuitive understanding of the underlying physics. If you do this, the solver will be featured in the classes in the following years 😊

Evaluation Criteria and Deliverables

Students must submit their project materials across two main components: a **Written Report** (compiled into a single PDF document of no more than 5 pages) and the **Source Code/Deployment Link** uploaded to the designated submission folder (to be set during the semester).

I. Written Report (Maximum 5 Pages): 50%

Please include the following sections in the report.

1. **Title:** Give a unique name to your project and simulator.
2. **Problem and Physics Description:** Clearly describe the core fluid mechanics principles and governing equations you chose to model. Explicitly define all independent and dependent variables, boundary conditions, and physical or geometric constraints imposed on the system. If the variables and constraints evolve over the course of development, please describe how they have evolved (which is closely related to Point 3 below).
3. **Prompt Engineering and Iteration History:** Document the step-by-step development of your simulation. Detail the exact natural language prompts used, the computational or visual glitches encountered, and how you refined your prompts to resolve these issues. For each major iteration, explain precisely how the change modified the logic, visual output, and/or performance of the simulator.
4. **Model Validation:** Quantitatively validate the accuracy of the simulator. Choose a set of representative values and perform a manual analytical calculation using your governing equations. Compare these theoretical results directly against the numerical outputs generated by your simulator to verify its validity.
5. **References:** State any references used to develop your ideas and simulators.

II. Simulation Model (Digital Submission): 50%

- **Code of Simulation Model:** A functional, interactive simulation tool that accurately calculates and displays fluid behaviours based on dynamic user inputs. Ideally, it should run directly in a web browser without requiring any additional plug-ins. HTML/JavaScript is the easiest option to this end. Other languages and/or platforms are acceptable as long as the resulting simulator can be hosted on the web.

The simulation model will be evaluated based on the following criteria:

1. **Scope:** Selection of a unique and interesting problem to simulate.
2. **Physics/Engineering:** Explicit mapping of governing equations to the simulation software.
3. **Interactivity:** Capability to dynamically vary key operational variables via the user interface.
4. **Visualization:** Clear, real-time rendering of key fluid phenomena (e.g., streamlines, particle tracking).
5. **Completeness:** Delivery of functional simulation software; developmental glitches are successfully addressed and fixed.
6. **User Experience:** An intuitive, user-friendly graphical user interface (GUI).
7. **Educational Impact:** Effectiveness in guiding users toward an intuitive understanding of core fluid mechanics concepts.

Note: As a group project, all group members receive the same base marks. However, individual grades may be modulated based on mandatory peer evaluations submitted at the end of the project.

Key Dates

- **Forming Groups:** May 31, 2026 (Week 3, Sunday) @ 11:59 pm
- **Project Submission:** August 2, 2026 (Week 12, Sunday) @ 11:59 pm

Final Remarks

- **Project Management:** The progress of the 1D project is self-managed by each group. The instructor is available throughout the semester to consult on simulator topics and discuss your project's progress. Please utilize scheduled office hours or arrange a specific time by appointment to discuss your 1D project workflow.
- **Examples:** Throughout the semester, the instructor will provide different examples of fluid simulators to ensure everyone has access to relevant references and inspiration.
- **Exhibition in Public Domain (Optional/Voluntary):** If you wish, you may opt to host your finalized simulator in a public repository. The instructor will set up this library later in the semester. With your permission, these interactive models will be shared with future cohorts across the world as open-access learning tools.
- Have a great vibe coding experience!

Last Updated: June 1, 2026